

Simulating Crude Oil Prices

Jacob Thomas

University of Utah – Spring 2026 – Advisor: Sean Lawley

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Abstract

A regime-switching Hawkes jump-diffusion model with Student- t innovations, selective skewing, and mean reversion is calibrated to 6,268 daily WTI crude oil log-returns (2000-2024). The ~ 47 -parameter model generates synthetic price histories matching historical excess kurtosis at 99.6% (26.36 vs. 26.45) and passing distributional goodness-of-fit tests at 95% confidence – tests the industry-standard Heston model fails. Retrospective 2025 forecasts are also validated against realized market outcomes.

1 Introduction

WTI crude oil daily returns exhibit three empirical properties that standard pricing models cannot capture simultaneously: (i) extreme fat tails with excess kurtosis ≈ 26 (Gaussian: 0), meaning extreme daily moves occur roughly five times more often than a normal distribution predicts; (ii) volatility clustering, where large moves tend to follow large moves; and (iii) jump dynamics – sudden, discontinuous price movements driven by geopolitical events, speculative trading, and other movers with 32 such events detected across 25 years of data.

Geometric Brownian Motion (GBM), the foundation of quantitative finance, assumes Gaussian returns, constant volatility, and independence – all of which are violated by oil prices. Extensions exist in the literature: regime-switching models allow the market to alternate between volatility states (1), Hawkes self-exciting processes capture the clustering of jumps (2), and fat-tailed distributions address the kurtosis gap. However, no prior work combines three-regime Markov switching, Hawkes self-exciting jumps, Student- t diffusion with selective skewing, mean reversion, and Lee-Mykland jump detection in one Monte Carlo framework for crude oil.

The model developed here fills that gap. The key results are: excess kurtosis match of 99.6% (26.36 simulated vs. 26.45 historical), annualized volatility match of 99.5%, 5% VaR match of 99.9%, distributional goodness-of-fit tests passed at 95% confidence, Kupiec VaR backtests passed, and 2025 retrospective forecasts validated against realized market outcomes.

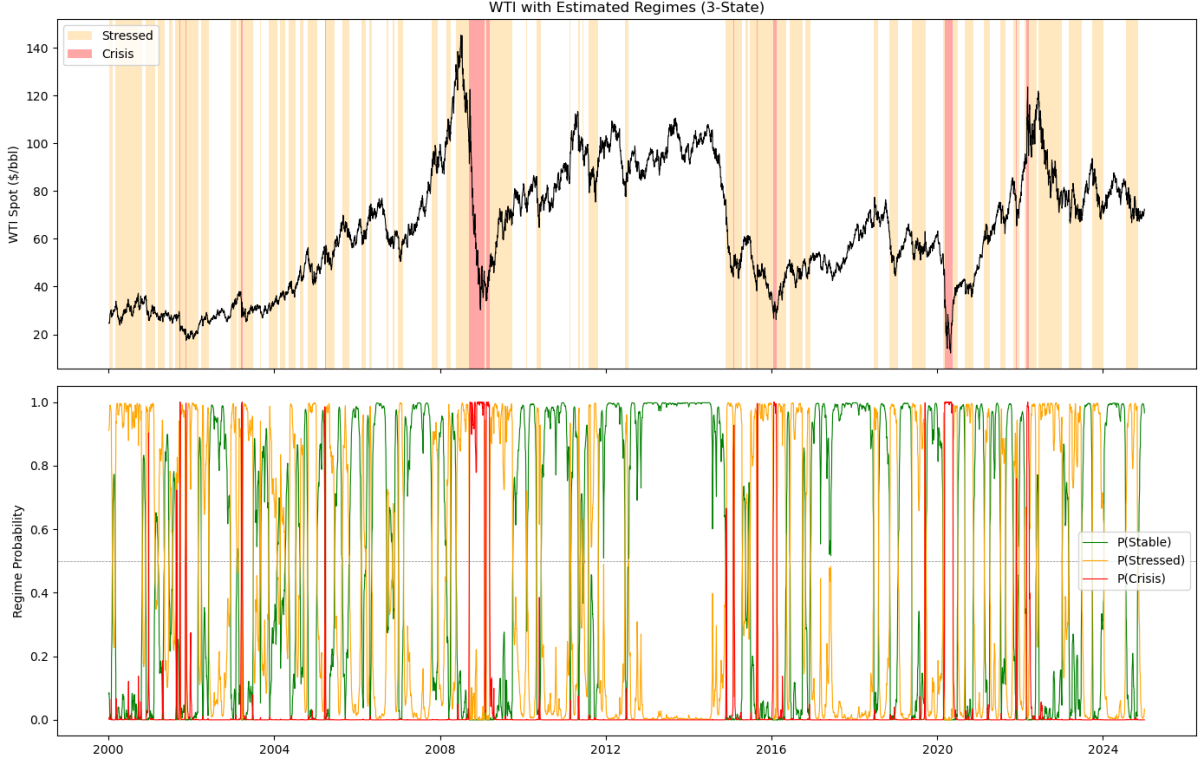


Figure 1: Historical WTI price series (2000–2024) with regime shading.

2 Data and Notation

Daily WTI spot prices from FRED (DCOILWTICO), January 2000–December 2024: 6,268 log-returns $r_t = \ln(S_t/S_{t-1})$. Key statistics: annualized volatility 43.5%, excess kurtosis 26.45, skewness +0.54, 5% VaR -3.91% .

Notation: S_t = price, $s_t \in \{1, 2, 3\}$ = regime (stable/stressed/crisis), μ_i, σ_i = drift and volatility, $P = [p_{ij}]$ = transition matrix, ν_i = Student- t df, a_i = skew parameter, κ_i, θ_i = mean reversion speed and level, $\bar{\lambda}_i, \alpha_i, \beta_i$ = Hawkes baseline/excitation/decay, λ_t = current intensity, J = jump size.

3 Model Architecture

The model is built incrementally, with each layer addressing a specific empirical failure:

Layer	Addresses	Kurtosis
GBM baseline	Drift + continuous noise	≈ 0
3-regime switching	Structural volatility breaks	$\approx 3\text{--}5$
Hawkes jumps	Clustered extreme events	$\approx 7\text{--}11$
Student- t diffusion	Fat-tailed daily returns	$\approx 21\text{--}23$
Df cap at 15	All regimes produce tails	≈ 26
Selective skew	Crisis asymmetry	Skewness improvement
Mean reversion	Equilibrium pull-back	Passes A-D test

Table 1: Layer progression and impact on excess kurtosis.

Approximately 99.5% of simulated days are ordinary diffusion days (no jump), so the shape of the diffusion innovation dominates the overall return distribution. Replacing Gaussian with Student- t on those 99.5% of days is the single highest-impact change.

The master equation:

$$\frac{dS_t}{S_t} = \mu_{s_t}(S_t) dt + \sigma_{s_t} dZ_t^{(\nu_{s_t})} + J_{s_t} dN_t, \quad s_t \in \{1, 2, 3\} \quad (1)$$

3.1 Three-Regime Markov Switching

The market occupies hidden states with distinct μ_i, σ_i . Transitions follow:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}, \quad \sum_j p_{ij} = 1 \quad \forall i \quad (2)$$

Regimes are estimated via the Hamilton filter and EM algorithm. The Hamilton filter predict step forecasts regime probabilities: $\hat{\xi}_{t|t-1}(j) = \sum_i p_{ij} \cdot \xi_{t-1}(i)$. The update step applies Bayes' rule after observing r_t :

$$\xi_t(j) = \frac{f(r_t | s_t = j) \cdot \hat{\xi}_{t|t-1}(j)}{\sum_k f(r_t | s_t = k) \cdot \hat{\xi}_{t|t-1}(k)} \quad (3)$$

where $f(r_t | s_t = j)$ is the Gaussian density with parameters μ_j, σ_j . The EM algorithm alternates: E-step runs the filter to assign regime probabilities; M-step re-estimates parameters via weighted statistics:

$$\hat{\mu}_j = \frac{\sum_t \xi_t(j) \cdot r_t}{\sum_t \xi_t(j)}, \quad \hat{\sigma}_j^2 = \frac{\sum_t \xi_t(j) \cdot (r_t - \hat{\mu}_j)^2}{\sum_t \xi_t(j)} \quad (4)$$

Iterated from 50 random starts, 300 iterations each.

Calibrated results:

Regime	Drift (μ)	Volatility (σ)	Annualized Vol	Duration
Stable	+0.001329	0.015308	24.3%	~ 47 days
Stressed	−0.000775	0.026915	42.7%	~ 30 days
Crisis	−0.002228	0.087000	138.1%	~ 13 days

Table 2: Calibrated regime parameters.

Calibrated transition matrix:

$$\hat{P} = \begin{bmatrix} 0.9788 & 0.0207 & 0.0005 \\ 0.0263 & 0.9668 & 0.0069 \\ 0.0000 & 0.0774 & 0.9226 \end{bmatrix} \quad (5)$$

$p_{31} \approx 0$: crises always de-escalate through the stressed regime first (novel finding, not imposed as a constraint).

3.2 Student- t Diffusion Innovations

Replace Gaussian with standardized Student- t :

$$dZ_t^{(\nu)} = \frac{T_\nu}{\sqrt{\nu/(\nu-2)}} \cdot \sqrt{dt} \quad (6)$$

Standardization forces variance to 1 so σ controls scale; fat tails are preserved. Per-regime ν_i fitted by maximum likelihood estimation (MLE), capped: $\nu_i = \min(\max(\hat{\nu}_i, 2.5), 15)$. Stable/stressed fitted at 24.85/45.67 (nearly Gaussian), capped to 15. Crisis: 6.46 (uncapped). The cap at 15 is an impactful decision – kurtosis jumped from ~ 21 to ~ 26 because 99.5% of days are ordinary diffusion days.

3.3 Selective Skewing

Applied only to crisis ($a_3 = +1.20$; $a_1 = a_2 = 0$):

$$u = \Phi\left(\frac{a_{s_t} \cdot T_\nu}{\sqrt{1 + a_{s_t}^2}}\right), \quad T_\nu^{\text{skewed}} = \begin{cases} +|T_\nu| & U \leq u \\ -|T_\nu| & U > u \end{cases} \quad (7)$$

Stable/stressed have negative within-regime skew; applying it would degrade overall skewness. The purpose is to improve the sign of skewness.

3.4 Mean-Reverting Drift

$$\mu_{s_t}(S_t) = \kappa_{s_t}(\theta_{s_t} - \ln S_t) \quad (8)$$

When price exceeds equilibrium ($\ln S_t > \theta$), drift is negative. Calibrated by regression: $r_t = a + b \cdot \ln S_{t-1} + \varepsilon$. If $b < 0$: $\kappa = -b \times 252$, $\theta = -a/b$. Theta is purely statistical – production cost is the economic intuition, not the calibration input. This enabled the Anderson-Darling test to pass ($p = 0.147$).

3.5 Lee-Mykland Jump Detection

Bipower variation (7) estimates local volatility robust to jumps:

$$\hat{\sigma}_t^{BV} = \sqrt{\frac{\pi}{2} \cdot \frac{1}{K-1} \sum_{j=1}^{K-1} |r_{t-j}| \cdot |r_{t-j-1}|} \quad (9)$$

Test statistic $L_t = r_t / \hat{\sigma}_t^{BV}$. Flag as jump if:

$$\frac{|L_t| - C_n}{S_n} > c_\alpha, \quad C_n = \sqrt{2 \ln n} - \frac{\ln \pi + \ln(\ln n)}{2\sqrt{2 \ln n}}, \quad S_n = \frac{1}{\sqrt{2 \ln n}} \quad (10)$$

At $\alpha = 0.05$, $K = 10$: 32 jumps detected across 6,268 days (3).

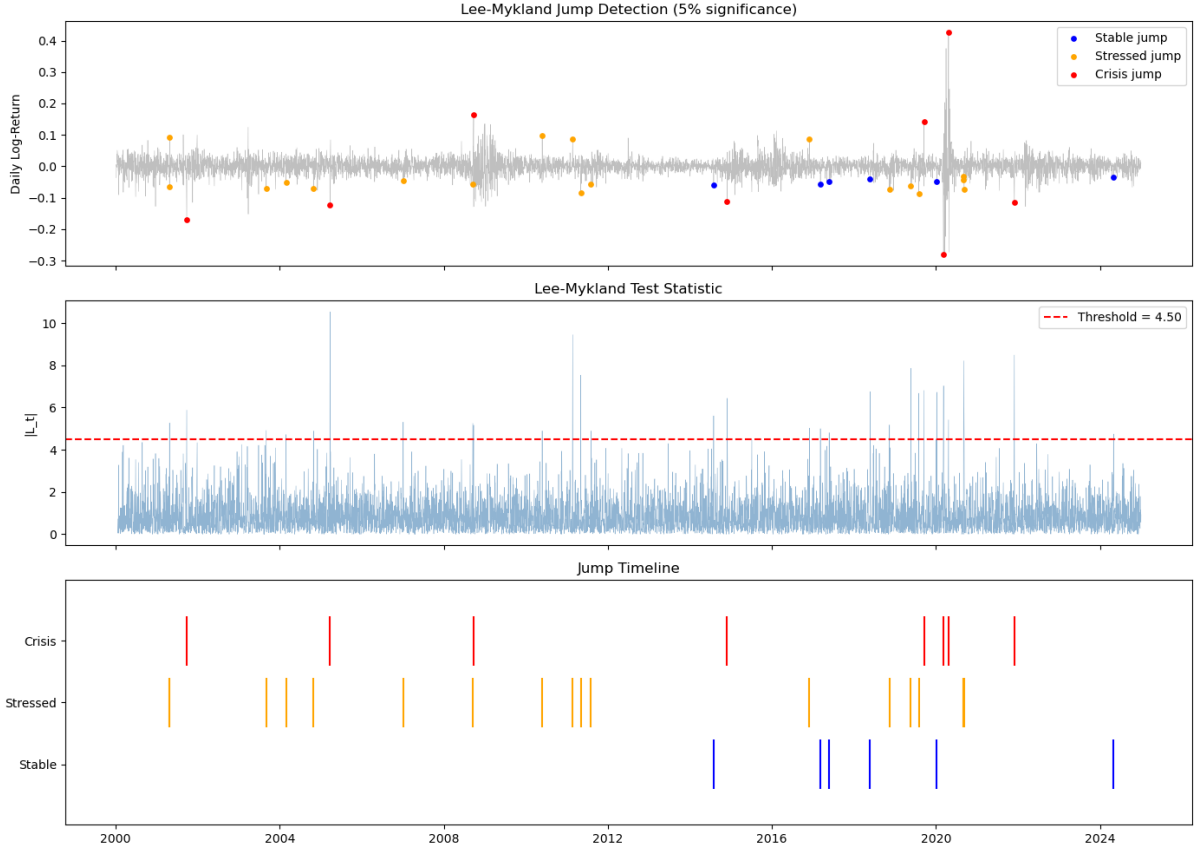


Figure 2: Returns with detected jumps, $|L_t|$ statistics with Gumbel threshold, and event timeline.

3.6 Hawkes Self-Exciting Jump Intensity

$$\lambda_t = \bar{\lambda}_{s_t} + (\lambda_{t-1} - \bar{\lambda}_{s_t})e^{-\beta_{s_t}\Delta t}, \quad p_{\text{jump}} = 1 - e^{-\lambda_t\Delta t} \quad (11)$$

If jump occurs: $\lambda_t \leftarrow \lambda_t + \alpha_{s_t}$. Branching ratio $n = \alpha/\beta < 1$. Estimated by MLE:

$$\ell = \sum_{i=1}^n \ln \lambda(t_i) - \bar{\lambda}T - \frac{\alpha}{\beta} \sum_{i=1}^n (1 - e^{-\beta(T-t_i)}) \quad (12)$$

with 20 random restarts. Crisis branching ratio: 17.6% ($\sim 4\times$ stable/stressed), meaning shocks cascade far more aggressively during panics.

Regime	$\bar{\lambda}$	α	β	Branching Ratio
Stable	0.003985	0.011981	0.304061	3.9%
Stressed	0.002139	0.008435	0.198920	4.2%
Crisis	0.001184	0.018592	0.105804	17.6%

Table 3: Calibrated Hawkes parameters by regime.

3.7 Asymmetric Jump Sizes

$$J_{s_t} = \begin{cases} +|T_{\nu^+}(0, \sigma_J^+)| & \text{prob } p^+ = 0.22 \\ -|T_{\nu^-}(0, \sigma_J^-)| & \text{prob } 1 - p^+ = 0.78 \end{cases} \quad (13)$$

Positive jumps are $\sim 2\times$ larger (mean +15.7%) but $\sim 3.6\times$ less frequent (mean negative: -7.9%).

3.8 Daily Price Update

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu_{s_t}(S_t) - \frac{\sigma_{s_t}^2}{2} \right) \Delta t + \sigma_{s_t} \cdot dZ_t^{(\nu_{s_t})} + J \cdot \mathbf{1}_{\text{jump}} \right] \quad (14)$$

Total: ~ 47 parameters from 6,268 observations (ratio 133:1).

3.9 Simulation Algorithm

For each of 5,000 paths, for each of 6,268 days:

1. **Regime transition:** Draw s_t from row s_{t-1} of P .
2. **Intensity decay:** $\lambda_t \leftarrow \bar{\lambda}_{s_t} + (\lambda_{t-1} - \bar{\lambda}_{s_t})e^{-\beta_{s_t}\Delta t}$
3. **Diffusion draw:** $T_\nu \sim t(\nu_{s_t})$. If crisis, apply skewing. Standardize: $dZ = T_\nu^{\text{skewed}} \cdot \sqrt{\Delta t(\nu - 2)/\nu}$.
4. **Mean-reverting drift:** $\mu_{mr} = \kappa(\theta - \ln S_{t-1})/252$ if $\kappa > 0$, else μ_{s_t} .

5. **Diffusion return:** $r_{\text{diff}} = (\mu_{mr} - \sigma^2/2)\Delta t + \sigma \cdot dZ$
6. **Jump check:** $p_{\text{jump}} = 1 - e^{-\lambda_t \Delta t}$; draw U ; if $U < p_{\text{jump}}$, jump occurs.
7. **If jump:** Draw from asymmetric Student- t . $r_{\text{total}} = r_{\text{diff}} + J$. Spike: $\lambda_t \leftarrow \lambda_t + \alpha_{st}$.
8. **Price update:** $S_{t+\Delta t} = S_t \exp(r_{\text{total}})$

The model generates synthetic oil prices day by day by computing percentage changes, then extracts log-returns from those prices and compares them against the 6,268 historical returns for validation.

3.10 Complete Parameter Set

Category	Parameters	Count
Regime drifts	μ_1, μ_2, μ_3	3
Regime volatilities	$\sigma_1, \sigma_2, \sigma_3$	3
Transition matrix	p_{ij} (6 free)	6
Diffusion df	ν_1, ν_2, ν_3	3
Diffusion skew	a_3 (crisis only)	1
Mean reversion speed	$\kappa_1, \kappa_2, \kappa_3$	3
Mean reversion level	$\theta_1, \theta_2, \theta_3$	3
Hawkes baselines	$\bar{\lambda}_1, \bar{\lambda}_2, \bar{\lambda}_3$	3
Hawkes excitation	$\alpha_1, \alpha_2, \alpha_3$	3
Hawkes decay	$\beta_1, \beta_2, \beta_3$	3
Jump sizes	df, loc, scale per direction per regime	≤ 18
Total		~ 47

Table 4: Complete parameter set. Observation-to-parameter ratio: $6,268/47 \approx 133$.

4 Results

4.1 Moment Comparison

Statistic	Historical	Simulated	Match
Annualized vol	43.47%	43.70%	99.5%
Excess kurtosis	26.45	26.36	99.6%
5% VaR	-3.91%	-3.92%	99.9%
1% VaR	-7.33%	-7.63%	95.9%
3-sigma exceedance	1.39%	1.45%	$\sim$ match
Skewness	+0.54	-0.49	Wrong sign

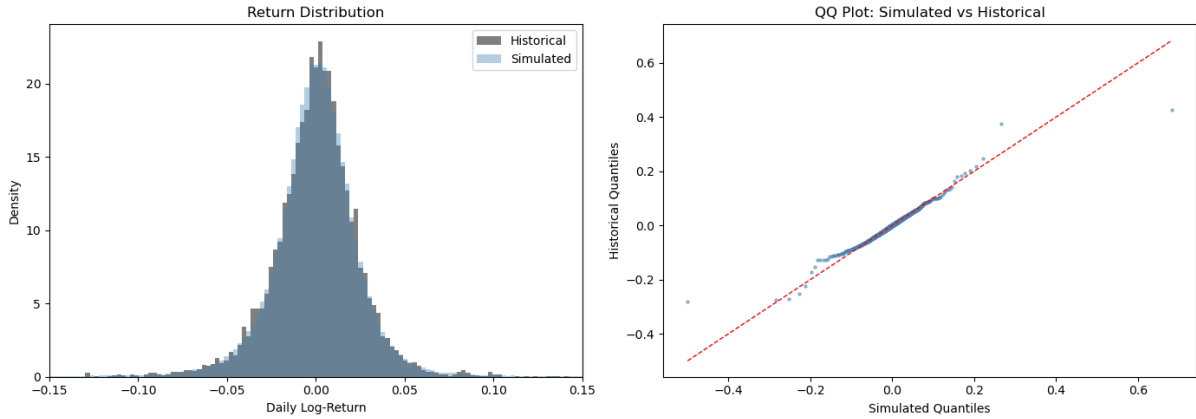


Figure 3: Histogram overlay of historical and simulated returns (left) and QQ plot (right).

4.2 Distributional Tests

Test	Statistic	p-value	Result
Standard KS	0.0219	0.100	PASS
Permutation KS (9,999)	0.0196	0.181	PASS
Anderson-Darling	0.8489	0.147	PASS
Permutation A-D (4,999)	-0.183	0.449	PASS

Kolmogorov-Smirnov (KS) measures the maximum CDF gap; Anderson-Darling up-weights tails. Permutation versions shuffle data into random groups thousands of times, computing p-values without asymptotic assumptions. Daniel et al. (4) showed the Heston model fails these tests on comparable data.

4.3 VaR Backtesting

VaR at level α is the loss exceeded on $\alpha\%$ of days. The Kupiec (5) likelihood ratio test checks whether the observed breach count matches the predicted rate

Test	Expected	Actual	Rate	p-value	Result
Kupiec 1%	62.7	56	0.89%	0.388	PASS
Kupiec 5%	313.4	312	4.98%	~ 1.0	PASS
Christoffersen 1%	—	—	—	0.0001	FAIL

The Christoffersen (6) fails because the unconditional threshold (-3.92%) is applied to all days. During a crisis (138% vol), this is breached repeatedly. The clustering reflects regime dynamics, not miscalibration.

4.4 Skewness Analysis

Historical +0.54 is driven by one observation – the 53% April 2020 rebound. Removing it: skewness drops to -0.05 . Per-path analysis (5,000 paths): 29.2% produce positive skewness, 90th percentile = +0.47. The model assigns $\sim 30\%$ probability to the historical outcome.

Remark 1. *Positive skewness in oil is a single-observation artifact, not a structural market property.*

4.5 OVX Validation

The Oil Volatility Index (OVX—not used in the model) validates jump detection: jump rates are $2.07\times$ higher when OVX exceeds its median. Mean OVX on jump days: 52.32 vs. 38.34 overall. Seven of 32 jumps map to named events (9/11, Lehman Brothers, Arab Spring, OPEC 2014, Saudi-Russia, COVID).

4.6 Overfitting Checks

Parameter ratio 133:1 (safe >30). Results stable across Lee-Mykland $K = 5\text{--}20$, df caps 8–30, training starts 2000–2008, and 5-fold cross-validation.

4.7 Forecast Validation (2025)

10,000 paths from $S_0 = \$72.44$, 2-year horizon, no fixed seed, same 47 parameters.

Prediction	Model	Actual	Assessment
6-mo median	\$68.76	$\sim \$65$	5.5% error, 25–50th pct
1-yr median	\$62.67	$\sim \$57.35$	8.5% error, 25–50th pct
Regime (6 mo)	53% stable, 43% stressed	Matched	Correct
1-yr drawdown	-39.6% median	$\sim -35\%$	Within 5 pts
1-yr P(loss)	67.8%	Lost 21%	Correct
Crisis in 2 yr	78.6%	Hormuz, ~ 15 mo	Confirmed

Additional applications (all from the same parameters): price distributions, time-to-target, jump risk (32% chance ≥ 1 jump/yr), hedging (95% put strike at \$33.80 for 1 yr), max drawdown (median -39.6% at 1 yr), and conditional forecasts (crisis vs. no-crisis scenarios).

5 Discussion

5.1 Significance

The model demonstrates that a tractable, interpretable parametric framework can capture crude oil return dynamics – outperforming the Heston stochastic volatility model on formal goodness-of-fit tests. Each layer addresses a specific empirical failure, and Table 1 quantifies exactly how much each one contributes. The same 47 parameters reproduce 25 years of historical behavior and generate the forward-looking distributions in Section 4.7. No separate forecasting model is required.

5.2 Limitations

- **Christoffersen failure:** VaR violations cluster due to unconditional thresholding. Regime-conditional VaR (different thresholds per regime) would address this.
- **Skewness sign:** Aggregate simulated skewness is negative, while historical is positive. For applications requiring accurate unconditional skewness, this is a limitation, though the per-path analysis demonstrates the model correctly assigns $\sim 30\%$ probability to the observed outcome.
- **Stationarity assumption:** Parameters are assumed stable over 25 years. Structural breaks (e.g., the shale revolution) could affect stability.
- **Single asset:** Cross-asset dynamics and macroeconomic factors are not modeled.

6 Conclusion

This paper presents a mean-reverting regime-switching Hawkes jump-diffusion model with Student- t innovations and selective skewing, calibrated to 25 years of WTI crude oil daily data. The ~ 47 -parameter model generates synthetic price paths that pass all four distributional goodness-of-fit tests at 95% confidence, match historical kurtosis at 99.6%, and produce forecasts validated against 2025 realized outcomes.

6.1 Novel Contributions

1. **First combined framework for oil:** Three-regime switching, Hawkes jumps, Student- t diffusion with selective skewing, mean reversion, and Lee-Mykland detection in one Monte Carlo framework.
2. **Passes Anderson-Darling where Heston fails:** The industry-standard stochastic volatility model fails distributional tests that this model passes.

3. **Crisis-to-stable ≈ 0 :** An empirical finding that crises always de-escalate through stressed, emerging naturally from the data.
4. **Skewness as sample artifact:** Per-path analysis demonstrates that historical positive skewness is driven by a single observation, not structural dynamics.
5. **OVX validates jump detection:** A $2.07\times$ higher jump rate during elevated implied volatility, using a fully independent data source.

6.2 Future Work

- **Multivariate Hawkes contagion:** Extend to jointly model oil, equities, bonds, and gold, capturing cross-asset shock transmission. This is the subject of a funded UROP research proposal.
- **Extended data:** Incorporate 2025-2026 data, including the Strait of Hormuz crisis, as a real-time stress test.
- **Regime-conditional VaR:** Address the Christoffersen failure by implementing separate thresholds per regime.

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